Arista Access Network QoS Congestion Event Resolution SOP (KR-ACC-56)

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5294

Description

The QoS congestion event on the Arista Access network, specifically on Node ID KR-ACC-56 located at KR Puram Access, is a critical failure domain that occurs due to the microbursts, interface tail drops, and QoS queue starvation. This phenomenon is a direct result of the interplay between the IEEE 802.1p/priority queuing and the Class-Based QoS (CBQoS) mechanisms implemented on the Arista 7250X series switch. The CBQoS policy maps the 802.1p/priority tags to specific queues, which are then scheduled using the Weighted Random Early Detection (WRED) algorithm. However, when the microbursts of high-priority traffic occur, it leads to a condition where the WRED algorithm prematurely drops packets, resulting in tail drops. This, in turn, causes the QoS queue to starve, leading to congestion. Furthermore, the CBQoS policy may not be configured correctly, leading to incorrect queue assignments and exacerbating the congestion. Therefore, it is essential to investigate and address the root causes of this issue to prevent further congestion and ensure high network uptime.

Pre-requisite Checks

- Verify that the redundant link is up and functioning correctly.

- Ensure that the remote hands team is dispatched to KR Puram Access for physical access.

- Backup the current configuration using the "show running-config" command.

Potential Root Causes

Cause	Probability
Incorrect CBQoS policy configuration	High
Insufficient buffer space on the interface	Medium
Hardware congestion due to microbursts	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy, Expected Output: Policy name:
KR-ACC-56-POLICY, Description: Verify the CBQoS policy configuration
using this command. The expected output will display the policy name,
which should match the node ID (KR-ACC-56).

Command: show interface | include buffer, Expected Output: Buffer
size: 131072, Description: Check the buffer size on the interface
using this command. The expected output will display the buffer size,
which should be sufficient to handle the traffic.

Command: show platform internal stats | grep WRED, Expected Output:
WRED drops: 0, Description: Verify the WRED drops using this command.
```

```
The expected output will display the WRED drops, which should be 0 to
indicate no premature packet drops.
```

Resolution Steps

Step 1: Step: Adjust CBQoS policy configuration, Command: config qos policy KR-ACC-56-POLICY queue-map 1 map-set 1 queue 1, Description: Adjust the CBQoS policy configuration to allocate sufficient queue space for high-priority traffic. This will prevent premature packet drops and tail drops.

Step 2: Step: Increase buffer space on the interface, Command: interface Ethernet1/1 buffer-size 262144, Description: Increase the buffer size on the interface to handle high traffic volumes. This will prevent buffer overflows and subsequent congestion.

Step 3: Step: Verify WRED configuration, Command: show platform internal stats | grep WRED, Description: Verify the WRED configuration to ensure it is properly set up to prevent premature packet drops.

Post-Check Verification

```
Command: show qos policy, Expected Output: Policy name:
KR-ACC-56-POLICY, Description: Verify the CBQoS policy configuration
after the resolution steps.

Command: show interface | include buffer, Expected Output: Buffer
size: 131072, Description: Verify the buffer size on the interface
after the resolution steps.

Command: show platform internal stats | grep WRED, Expected Output:
WRED drops: 0, Description: Verify the WRED drops after the resolution
steps.
```

Roll-back Procedures

- Step: Restore original CBQoS policy configuration, Command: config qos policy KR-ACC-56-POLICY restore, Description: Restore the original CBQoS policy configuration if the adjustments made during the resolution steps fail to resolve the issue.

- Step: Revert buffer size change, Command: interface Ethernet1/1 buffer-size 131072, Description: Revert the buffer size change if the increase in buffer space during the resolution steps fails to resolve the issue.